

Revised Generation Interconnection System Impact Study Report

For Queue Project Q20-1372

Kaukauna North 138 kV
April 16, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1372. The Interconnection Customer proposes a 99.0 MW battery energy storage facility interconnecting at the Kaukauna North 138 kV substation in the Pennsylvania area. The request was received on January 01, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 466 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$3,017,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Pehastin - Statesville Highway 64 138kV Line	Ft Gibson - Armstrong 500kV	105.1	249	5.2
Statesville Highway 64 - Oneill 138kV	Armsrong Substation - Ft Gibson 500kV Line	104.2	243	5.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Statesville Highway 64 - Peshastin 138kV	Reconductor line	\$1,703,000	100% (\$1,703,000)
Statesville Highway 64 - Oneill 138kV	Reconductor line	\$1,314,000	100% (\$1,314,000)
	Total allocated to project		\$3,017,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.